REDUNDANT ECOG COVERAGE ON FOLDED SURFACES: DESIGN BOUNDS AND CERTIFIED MESH COVERAGE

ADITI BOSE, DEEP BHATTACHARJEE, AND USHASHI BHATTACHARYA

Abstract. Cortical electrode arrays must cover a target despite contact failure and surface folding. Under a common-radius footprint model, we formulate this requirement through the k-th covering radius and prove its equivalence to coverage after every set of $k-1$ contact failures. We distinguish coverage of sampled vertices, coverage under a finite triangle rule, and coverage of every point of the represented piecewise-flat surface. For the last problem, we prove the sufficient face test $m + D/\sqrt{3} < r$, where m is the largest vertex k-th-distance upper bound and D the longest edge; its universal triangle constant is sharp. Valid distance bounds are constructed from upward-rounded edge lengths and integer shortest-path sums, and the final face comparisons use exact rational arithmetic. On the 3,777 mm^2 colin27 target, the documented 64-contact grid leaves 55.1% unseen under the existing triangle rule, whereas a stored 66-contact design passes that rule at 8 mm. The stronger continuous-face test certifies all 9,659 target triangles with 80 contacts at 8 mm for single coverage and 160 contacts at 8 mm for double coverage. These are sufficient certificates for stored designs, not minimum-contact results. We also analyze site restrictions, onset-zone capture and random failures in the finite graph model. The certificates depend on the represented geometry and footprint model; no patient outcome or anatomical error bound is established.

1. Introduction

Epilepsy is drug-resistant when two appropriately chosen and tolerated antiseizure medications have failed [22], and a substantial minority of patients reach that point [23]. For focal drug-resistant epilepsy, resective surgery can end the seizures [21]. For temporal lobe epilepsy, randomised trials found it better than continued medical therapy [34, 10]. Its stated goal is freedom from disabling seizures (Engel class I) or from all seizures (ILAE class 1) [11, 35, 21]. Surgery aims at the epileptogenic zone, the cortex whose removal is needed for seizure freedom, and when intracranial recording is needed, a central estimate of it is the seizure onset zone that recording shows [28].

When the non-invasive evaluation (semiology, scalp video-EEG, MRI, PET, SPECT, MEG) does not settle where seizures begin, contacts are implanted: subdural grids and strips for electrocorticography (ECoG), or depth leads for stereo-EEG [20, 27]. The implantation is hypothesis driven. It samples a territory X that the evaluation says should contain the onset zone, and what it does not sample it cannot localise (Fig. 1). An onset zone that falls between contacts is missed or truncated, and the resection is then planned around a false or partial picture, or not at all. Invasive monitoring carries risk: for subdural grids, complications have been associated with the number of electrodes [15], and stereo-EEG has its own complication profile [25]. The number of contacts cannot simply be raised until nothing is missed.

This paper asks a narrow question and answers it exactly on the triangulated surface: given the patient's cortical surface, the territory X, and a radius r within which a contact is credited with recording an onset, where should n contacts go so that every point of X is seen at least k times, and how small can n be? The answer turns on one number, the k-th covering radius of X measured along the folded surface (Section 2). It gives three things a clinician can use before implantation.

Date: October 28, 2026.

Key words and phrases. electrocorticography, epilepsy surgery, seizure onset zone, electrode placement, covering radius, k-fold coverage, nerve theorem.

Deep Bhattacharjee is the corresponding author.

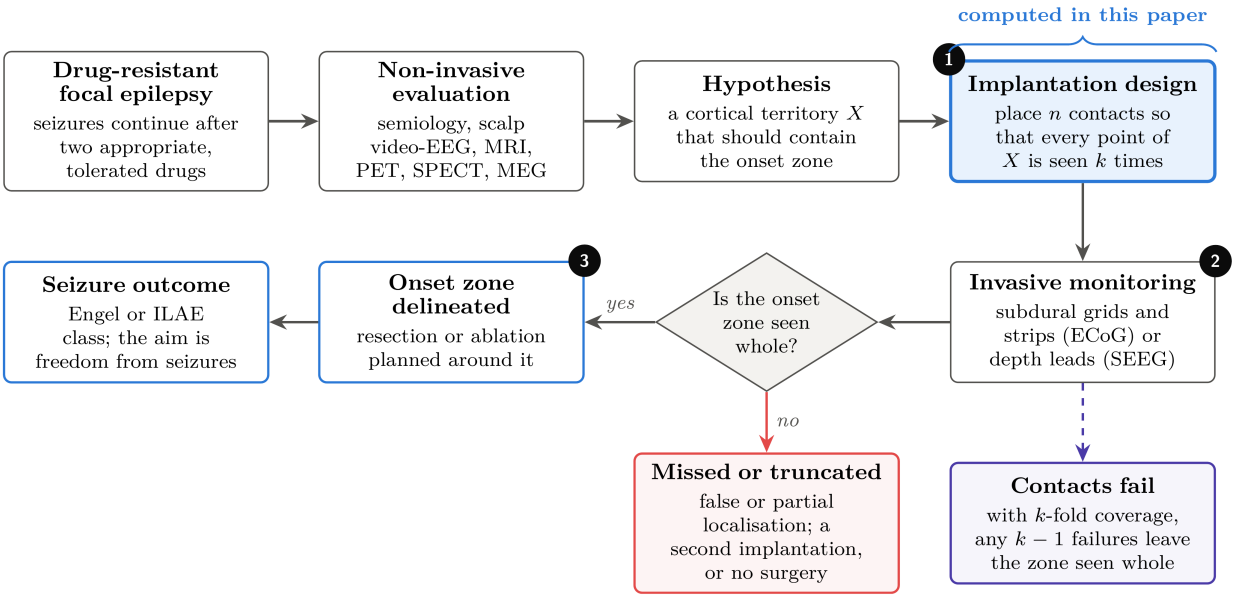


FIGURE 1. Where contact placement enters the surgical pathway. The implantation is designed from a hypothesis, a cortical territory X expected to contain the seizure onset zone (step 1). What the invasive recording can localise is limited to what its contacts see (step 2), and a resection aimed at seizure freedom is planned around the onset zone the recording delineates (step 3). An onset zone that is missed or seen only in part leads to false or partial localisation. With k-fold coverage, any $k-1$ contact failures during monitoring leave every onset zone in X seen whole (Proposition 3). This paper computes step 1.

First, a guarantee: when the covering radius is below r, an onset zone of any size and position in X is seen whole, and stays seen whole after any $k-1$ contacts fail. Second, a design rule that meets the guarantee; for single coverage with sites anywhere on X it is provably within a factor of two of the best covering radius for its contact count, up to one mesh edge. Third, an obstruction: contacts confined to the gyral crowns, as on a subdural sheet, cannot meet the guarantee below a radius that is fixed by the folding and not by the number of contacts.

We compute all of this on the colin27 template brain [17, 26], where a documented clinical grid layout [33] can be compared with designed ones, and we extend the coverage question to the one a surgeon asks: for an onset zone of radius s placed anywhere in X, how often is it seen whole, how often is it missed, and how often does it survive contact failures (Section 5)?

What this paper does not claim. Coverage is geometry. Whether a patient becomes seizure-free depends on much more: whether the onset zone recorded is the epileptogenic zone, whether it can be resected without unacceptable deficit, and whether the epilepsy is focal or network-driven [32, 1]. Complete coverage of the territory is what lets a resection planned on the recording contain the onset zone by design, not a promise of cure, and incomplete resection of the recorded onset zone has itself been reported not to predict outcome in one cohort [13]. No patient data and no outcome are used here. Section 6 states the protocol that would test whether optimal coverage changes outcome.

Relation to earlier work. Trajectory planning for stereo-EEG optimises safety and sampling of target structures [6, 31], electrode spacing has been studied for how much signal an array resolves [30], and Jaber et al. test after the fact whether an implantation sampled the epileptogenic zone [19]. Coverage by sensors with a sensing radius is classical in sensor networks [18], its topology

can be read from the nerve of the footprints [3, 8], and the k-fold region from the multicover nerve [29]; the certificate used here as a check follows [2]. The placement rule for unrestricted sites is farthest-point sampling, the greedy algorithm for the k-centre problem [14, 16]. What is new is the statement of ECoG placement as a covering-radius problem on the folded surface, the site floor that it implies for subdural sheets, and the onset-zone capture and failure analysis of designed arrays. The combinatorial content of the propositions on the mesh is also proved in the Lean 4 proof assistant [7] (Section 7).

2. Coverage of a territory, and of an onset zone

Let M be the patient's pial surface, a closed surface with the geodesic distance d (the length of the shortest path along the surface). Let $X \subseteq M$ be the target territory, a compact region; $C \subseteq M$ the sites a contact may occupy; and $P = \{p_1, \dots, p_n\} \subseteq C$ the contacts. A contact sees the points within distance r of it along the surface; its footprint is $B_i = \{x \in M : \mathrm{d}(x, p_i) < r\}$. The multiplicity of a point is $m(x) = \#\{i : x \in B_i\}$, and the region seen at least k times is $R_k = \{x : m(x) \geq k\}$, so that $R_1 \supseteq R_2 \supseteq \cdots$ (Fig. 3).

Definition 1. For $x \in M$ let $\mathrm{d}_k(x, P)$ be the distance from x to its k-th nearest contact. The *k-th covering radius* of X by P is

$$\rho_k(P) = \max_{x \in X} \mathrm{d}_k(x, P).$$

Proposition 2 (coverage is a covering radius)**.** *$X \subseteq R_k$ if and only if $\rho_k(P) < r$.*

Proof. A point x lies in R_k exactly when at least k contacts are within distance less than r of it, that is, when $\mathrm{d}_k(x, P) < r$. This holds for every $x \in X$ exactly when it holds for a maximiser of $\mathrm{d}_k(\cdot, P)$ on X, which exists because $\mathrm{d}_k(\cdot, P)$ is continuous and X is compact. □

So the whole design problem is to push $\rho_k(P)$ below r with as few contacts as possible.

Onset zones and contact failure. An onset zone is a region $Z \subseteq X$ of unknown position. We say Z is *seen whole* if $Z \subseteq R_1$, and *missed* if $Z \cap R_1 = \emptyset$. Contacts fail during monitoring, through lost contact with the cortex, noise or artefact, and the question is whether Z is still seen whole by the contacts that remain.

Proposition 3 (failure-proof sampling)**.** *Let $Z \subseteq X$ and $q \geq 0$. Every set of at most q failed contacts leaves Z seen whole by the survivors if and only if $Z \subseteq R_{q+1}$. In particular, if $\rho_{q+1}(P) < r$, every onset zone in X, of any size and position, is seen whole after any q contacts fail.*

Proof. If $Z \subseteq R_{q+1}$, a point $x \in Z$ lies in at least $q+1$ footprints, at most q of which have failed, so a surviving footprint contains x. Conversely, if some $x \in Z$ has $m(x) \leq q$, let the $m(x)$ contacts whose footprints contain x fail; that is at most q failures, and x is no longer seen. The last sentence follows from Proposition 2 with $k = q + 1$. □

The guarantee is worst case: it holds for every failure set, however unlucky, not merely on average. Section 5 also measures the average under random failure.

Capture of an onset zone of given size. When $X \not\subseteq R_k$ the answer depends on the size and position of the onset zone. Model it as the closed geodesic disk $Z_s(x) = \{y : \mathrm{d}(x, y) \leq s\}$ of radius s about a centre x, admissible when $Z_s(x) \subseteq X$, and let A_s be the set of admissible centres. With μ the area measure, define, for every s with $\mu(A_s) > 0$,

$$W_k(s) = \frac{\mu\{x \in A_s : Z_s(x) \subseteq R_k\}}{\mu(A_s)}, \qquad L(s) = \frac{\mu\{x \in A_s : Z_s(x) \cap R_1 = \emptyset\}}{\mu(A_s)}.$$

If the centre of the onset zone is equally likely to be anywhere it fits in X, $W_1(s)$ is the probability that the array sees it whole, $W_2(s)$ the probability that it still does after any one failure, and $L(s)$

the probability that it is missed entirely. The uniform prior is an assumption of the model. With a patient-specific prior, for instance from the concordance of non-invasive tests, μ is replaced by that prior and nothing else changes.

Proposition 4 (capture by erosion)**.** *$Z_s(x) \subseteq R_k$ if and only if $\mathrm{d}(x, M \setminus R_k) > s$. For $s > 0$, $Z_s(x) \cap R_1 = \emptyset$ if and only if $\mathrm{d}(x, R_1) \geq s$; for $s = 0$, if and only if $x \notin R_1$. Consequently, for every s with $\mu(A_s) > 0$:*

1. *if $X \subseteq R_k$ then $W_k(s) = 1$ and $L(s) = 0$, and $W_k(0) = \mu(X \cap R_k)/\mu(X)$ in general;*
2. *with $h(P) = \max_{x \in X} \min\{\mathrm{d}(x, R_1), \mathrm{d}(x, M \setminus X)\}$, $L(s) > 0$ if $s < h(P)$ and $L(s) = 0$ if $s > h(P)$: $h(P)$ is the radius of the largest onset zone the array can miss entirely. Moreover*

$$h(P) \;\geq\; \max_{x \in X} \min\{\mathrm{d}_1(x, P) - r,\; \mathrm{d}(x, M \setminus X)\}.$$

Proof. The set $M \setminus R_k$ is closed, so it avoids the closed disk $Z_s(x)$ exactly when its distance from x exceeds s. The set R_1 is open. Let $s > 0$: if $\mathrm{d}(x, R_1) \geq s$ and some $y \in R_1$ had $\mathrm{d}(x, y) \leq s$, then $\mathrm{d}(x, y) = s > 0$, and points of a shortest path from x to y close to y would lie in R_1 at distance less than s; so $Z_s(x) \cap R_1 = \emptyset$ exactly when $\mathrm{d}(x, R_1) \geq s$. The same argument with the open set $M \setminus X$ shows that, for $s > 0$, x is admissible exactly when $\mathrm{d}(x, M \setminus X) \geq s$, while $A_0 = X$. For (1), if $X \subseteq R_k$ every admissible disk lies in R_k and meets R_1; and $A_0 = X$, $Z_0(x) = \{x\}$. For (2), if $s < h(P)$ the set of x with $\mathrm{d}(x, R_1) > s$ and $\mathrm{d}(x, M \setminus X) > s$ is open, by continuity of both distances, and nonempty, so it has positive area, and each of its points is an admissible centre whose disk is missed; if $s > h(P)$ no admissible centre has $\mathrm{d}(x, R_1) \geq s$. For the bound, if $y \in R_1$ then $\mathrm{d}(y, p) < r$ for some contact p, so $\mathrm{d}(x, y) \geq \mathrm{d}(x, p) - r \geq \mathrm{d}_1(x, P) - r$. □

Each of W_k and L is therefore one multi-source Dijkstra run [9], from the unseen set or from the seen set, followed by a threshold at each s. Part (2) turns the covering radius into a clinical statement: an array whose covering radius exceeds the footprint by δ at a point at least δ inside X can miss an onset zone of radius about δ there entirely, whatever else it records.

On the triangulated surface. Every computation is done on the triangle mesh of the pial surface, and the definitions have to be carried over in a way that keeps the propositions exact. We take triangles as the atoms. Distances are shortest paths on the edge graph of the mesh. The distance from a contact p to a triangle T is the distance to its farthest vertex, $e(p, T) = \max_{v \in T} \mathrm{d}(v, p)$, and T lies in the footprint of p when $e(p, T) < r$, so that a triangle counts as seen only when a single contact sees all of it. R_k is the set of triangles in at least k footprints, and the k-th covering radius is $\rho_k(P) = \max_{T \in X}\big(k\text{-th smallest of } e(p, T) \text{ over } p \in P\big)$. The onset zone about a triangle T_0 of X is the set of triangles whose centroids lie within distance s of the centroid of T_0, distances now running on the edge graph with every centroid joined to the vertices of its triangle; μ is triangle area. With these definitions Propositions 2–4 hold on the mesh with the same proofs, reading triangles for points, both conditions of Proposition 4 as strict inequalities between finite distances, and $W_k(0) = \mu(X \cap R_k)/\mu(X)$ exactly; the lower bound on $h(P)$ is a statement about the continuous surface.

The triangle rule is more restrictive than the vertex rule, and the choice matters in practice. The vertex covering radius $\rho_k^{\mathrm{v}}(P) = \max_{v \in X} \mathrm{d}_k(v, P)$, taken over the vertices of X alone, satisfies

$$\rho_k^{\mathrm{v}}(P) \;\leq\; \rho_k(P) \;\leq\; \rho_k^{\mathrm{v}}(P) + \ell,$$

where ℓ is the longest edge of X (2.96 mm here, median 0.98 mm), because $e(p, T) \leq \mathrm{d}(v, p) + \ell$ for every vertex v of T. It can therefore fall below r while a triangle is still unseen: 64 contacts placed by farthest-point sampling have $\rho_1^{\mathrm{v}} = 6.99$ mm but $\rho_1 = 8.12$ mm, and at $r = 8$ mm they leave 1.0 mm^2 unseen (Section 5).

The numerical site floors and onset-zone capture quantities below refer to this finite graph and triangle-rule model. Edge-graph distances are upper bounds on intrinsic surface distances, so a

numerical graph site floor alone does not prove an obstruction at the same radius for continuous intrinsic footprints. The following theorem gives separate positive certificates for those footprints.

A certificate for every point of the represented surface. The triangle rule defines a finite coverage problem. Its threshold tests do not, by themselves, certify the intrinsic distance of every interior point of a face. In particular, intrinsic balls on a folded surface need not be convex. We give a separate sufficient certificate for the continuous piecewise-flat surface represented by the mesh. This certificate permits different contacts to witness coverage at different vertices and points.

Let $u_i(v)$ be a proved upper bound on the intrinsic mesh distance $\mathrm{d}(p_i, v)$, with contacts at mesh vertices. Write $u_k(v)$ for the k-th smallest of these bounds. For a target triangle T, let D_T be its longest Euclidean edge and $m_T = \max_{v\in\mathrm{vert}(T)} u_k(v)$.

Theorem 5 (whole-face coverage)**.** *Every point of T belongs to at least k open radius-r intrinsic balls if*

$$m_T + D_T/\sqrt{3} < r.$$

For closed balls the non-strict inequality suffices. If this holds for every target face, the whole represented target remains covered after any $k-1$ contacts fail. The constant $1/\sqrt{3}$ is the smallest universal constant possible when only the vertex bounds and longest edge are used.

Proof. Every function $\mathrm{d}(p_i, \cdot)$ is 1-Lipschitz. Their k-th order statistic g_k is also 1-Lipschitz: moving a point by h increases each of the k distances below the old k-th distance by at most h, and reversing the points gives the opposite inequality. Coordinatewise upper bounds preserve order statistics, so $g_k(v) \le u_k(v)$.

For $x = \sum_{j=1}^{3} \lambda_j v_j \in T$, the barycentric variance identity gives

$$\sum_j \lambda_j |x - v_j|^2 = \sum_{i<j} \lambda_i \lambda_j |v_i - v_j|^2 \le \frac{D_T^2}{2}\left(1 - \sum_j \lambda_j^2\right) \le D_T^2/3.$$

At least one vertex therefore has $|x - v_j| \le D_T/\sqrt{3}$. The straight segment inside the face is a permissible surface path, so $g_k(x) \le g_k(v_j) + \mathrm{d}(v_j, x) \le m_T + D_T/\sqrt{3}$. This proves both coverage assertions. At each point at least k distinct contacts cover it; deleting $k-1$ leaves a survivor.

For sharpness, take $k = 1$ and put contacts at the vertices of an equilateral triangle of side D. The nearest-contact distance is zero at all vertices and $D/\sqrt{3}$ at the centre. Any smaller constant would certify a radius strictly below the centre's distance. □

Corollary 6 (bounded location and radius error)**.** *Suppose each actual contact lies within ε of its nominal location in the same intrinsic metric, and its actual footprint radius is at least $r - \delta$, where $\varepsilon, \delta \ge 0$. The sufficient open-ball test becomes $m_T + D_T/\sqrt{3} + \varepsilon + \delta < r$.*

Proof. The triangle inequality increases each contact distance by at most ε; decreasing every radius by at most δ gives the stated margin. Apply Theorem 5. □

Constructing genuine distance upper bounds. We obtain these bounds from explicit mesh-edge paths. Treat the stored binary64 coordinates as exact dyadic numbers and fix a length quantum $h = 2^{-20}$ mm. For every edge with exact squared length s_e, compute

$$w_e = \left\lceil \sqrt{s_e/h^2} \right\rceil$$

using integer arithmetic and an integer square root. Its surface length is at most hw_e. Dijkstra's algorithm with these integer weights returns a path whose length bound is the exact number $h\sum_e w_e$. Conversion to binary64 is rounded outward, with an exact rational check. These numbers are therefore upper bounds on intrinsic mesh distances, even though the shortest intrinsic path can cross face interiors and be shorter.

TABLE 1. Whole-face certificates using upward path bounds. Every entry certifies all 9659 target triangles as continuous piecewise-flat faces. Contact sets are prefixes of the stored free-contact sequence; these sufficient examples do not establish globally minimum counts.

Contacts	Radius (mm)	Multiplicity k	Failures tolerated
64	9	1	0
80	8	1	0
112	10	2	1
128	9	2	1
160	8	2	1

The search is truncated at 16 mm for efficiency. An unreached vertex is assigned the conservative bound $(V-1)h\max_e w_e$, justified by a simple path in the connected V-vertex edge graph. It is never assigned zero or declared covered because it was unreached. Finally, putting $t = r - m_T$, the whole-face decision is the exact rational comparison $t > 0$ and $D_T^2 < 3t^2$; closed-ball coverage uses non-strict signs. An unresolved face may already be covered and is not an uncovered-point certificate.

The target is the stored list of 9659 faces selected by the earlier patch construction. We certify their union; the floating-point selection of that list is not a certificate that it equals an exact 35 mm intrinsic disk. Likewise, a certificate on this mesh does not bound discrepancies between the template and a patient's cortex.

Table 1 and Fig. 2 report the new computation. The machine-readable audit record in the repository contains exact input hashes. The displayed contact counts are drawn from the prescribed set $64, 66, 80, 96, 112, 128, 160$; no claim is made about untested intermediate counts. Their certificates are independent of the good-cover hypothesis needed for the nerve theorem. The previously reported triangle-rule results, such as 66 contacts at 8 mm, remain results of that finite rule and are not promoted to the stronger whole-face guarantee.

The shape of the covered region. Where coverage is not complete, its shape matters. A hole in R_1, a pocket of unseen cortex surrounded by seen cortex, is exactly where an onset zone can hide while every contact around it records (Fig. 3a). The topology of R_k can be read from which footprints overlap: when every nonempty intersection of footprints is contractible, R_k is homotopy equivalent to the k-shadow of the nerve of the footprints [29, 2], whose Betti numbers (b_0, b_1) count pieces and holes. We use this certificate as an independent check on every design, on the understanding established in earlier work [4]: geodesic footprints on a folded cortex are not always convex, so the certificate is checked against a direct computation on the mesh, and a certificate of $(1, 0)$ says that R_k is one piece without holes, not that it contains X. Containment is decided by Proposition 2.

3. Site constraints and coverage obstructions

A subdural grid is a flat sheet laid on the surface of the brain. Its contacts sit on the gyral crowns; the banks and fundi of the sulci can lie a centimetre or more of cortex away along the surface, however close they are in a straight line (Fig. 4). Three elementary facts make this exact and say what can be done instead. The first holds for any k; take C finite (on a mesh, a set of vertices) and P a set of distinct sites of C.

Proposition 7 (site floor)**.** *Put $\rho_C^{(k)} = \max_{x\in X} \mathrm{d}_k(x, C)$, the distance from x to its k-th nearest site of C, maximised over X, and $\rho_C = \rho_C^{(1)}$. Then $\rho_k(P) \geq \rho_C^{(k)}$ for every $P \subseteq C$. So if $r \leq \rho_C^{(k)}$, no array with its contacts in C sees all of X k times, at any number of contacts.*

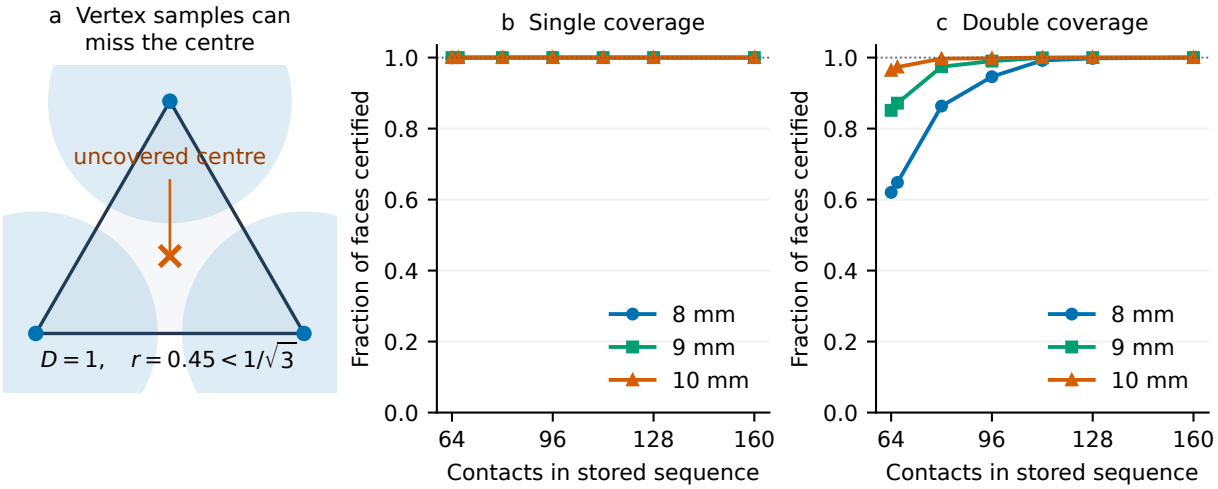


FIGURE 2. From vertex samples to whole-face coverage. (a) Three contacts at the vertices of a unit equilateral triangle cover all vertices at radius 0.45, while leaving its centre uncovered. The sharp universal vertex margin is $D/\sqrt{3}$. (b,c) Fraction of the template target faces passing the rigorous test for single and double coverage. A value of one certifies every point of the represented target. Lower fractions indicate unresolved faces, not a measured uncovered area.

Proof. Since $P \subseteq C$, the k nearest contacts of any x are among its sites in C, so $\mathrm{d}_k(x,P) \geq \mathrm{d}_k(x,C)$; take x attaining $\rho_C^{(k)}$. The second sentence is Proposition 2. On the mesh the same argument applies to $e(\cdot,T)$ triangle by triangle. □

Proposition 8 (packing bound)**.** *If $S \subseteq X$ has $\mathrm{d}(s,s') \geq 2r$ for all distinct $s,s' \in S$, then every array with $X \subseteq R_1$ has at least $|S|$ contacts.*

Proof. If s and s' lay in one footprint $B(p,r)$, then $\mathrm{d}(s,s') \leq \mathrm{d}(s,p)+\mathrm{d}(p,s') < 2r$. So each footprint contains at most one point of S, and the footprints cover S. On the mesh take S a set of vertices of X: each lies in a seen triangle and so within r of the contact that sees it. □

Proposition 9 (farthest-point placement is within a factor two)**.** *On a mesh, let $p_1, p_2, \ldots$ be vertices of X, each p_{j+1} a vertex of X farthest from $P_j = \{p_1,\ldots,p_j\}$. Then for every n and every set Q of n vertices of M, $\rho_1^{\mathrm{v}}(P_n) \leq 2\rho_1^{\mathrm{v}}(Q)$, and $\rho_1(P_n) \leq 2\rho_1(Q) + \ell$.*

Proof. Since p_{j+1} is a farthest vertex of X from P_j, $\mathrm{d}(p_{j+1},P_j) = \rho_1^{\mathrm{v}}(P_j)$, and $\rho_1^{\mathrm{v}}(P_j)$ does not increase with j. So any two of $p_1,\ldots,p_{n+1}$ are at least $\rho_1^{\mathrm{v}}(P_n)$ apart. Send each of these $n+1$ points to its nearest point of Q; two of them, p_i and p_j, share a nearest point q, and $\rho_1^{\mathrm{v}}(P_n) \leq \mathrm{d}(p_i,p_j) \leq \mathrm{d}(p_i,q) + \mathrm{d}(q,p_j) \leq 2\rho_1^{\mathrm{v}}(Q)$. The second inequality follows from the first and the bounds between ρ_1^{v} and ρ_1. □

Proposition 7 is the obstruction. Proposition 8 bounds the contact count from below, and Proposition 9 says the placement rule we use loses at most a factor two in radius, up to one mesh edge, against any placement with the same number of contacts. On a surface the count scales like the inverse square of the radius, so the factor two in radius becomes about four in contact count. The guarantee is for single coverage with sites anywhere on X. For $k \geq 2$ the same rule, and for sites restricted to C the worst-triangle rule of Section 4, are heuristics, and their designs are checked by Proposition 2 rather than bounded in advance.

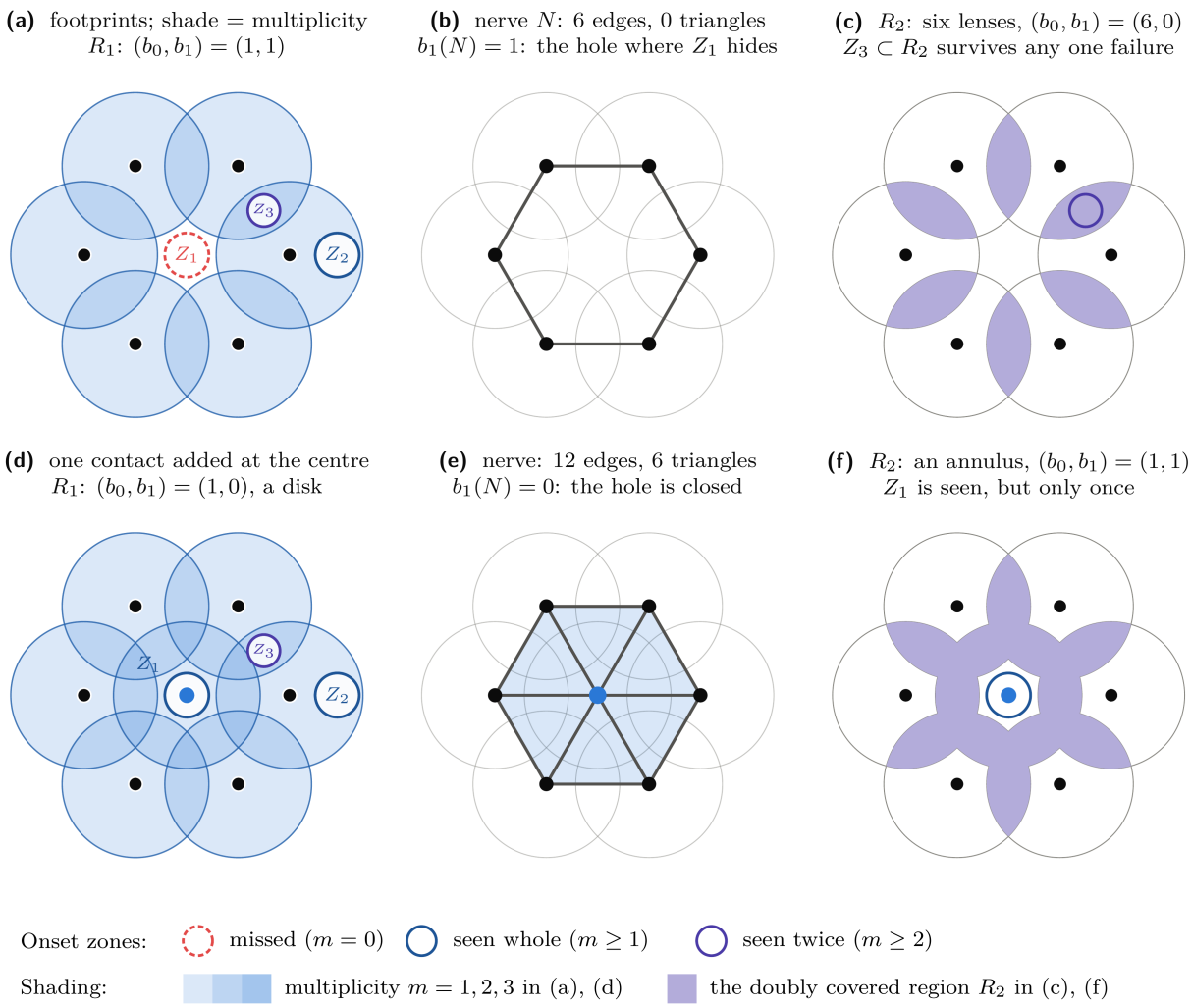


FIGURE 3. Footprints, the nerve, and what they say about an onset zone. Six contacts on a ring, footprints of radius 1 (top), and the same with one contact added at the centre (bottom). (a) The multiplicity m of the footprints. R_1 has a hole, and an onset zone Z_1 inside it is missed although six contacts surround it; Z_2 is seen whole once; Z_3 lies where two footprints overlap. (b) The nerve: an edge for each pair of overlapping footprints, a triangle for each triple with a common point. Its cycle is the hole of (a), found from overlaps alone. (c) The doubly covered region R_2, six lenses; $Z_3 \subset R_2$ is still seen whole if any one contact fails (Proposition 3). (d)–(f) One contact added at the centre closes the hole: Z_1 is now seen, but only once, since R_2 is an annulus around it. Every statement in the panel titles was checked numerically.

4. Designing the array

The design procedure follows from Section 2 and is summarised in Fig. 5. Given M, X, C, r and k: compute the site floor $\rho_C^{(k)}$ over every admissible site that can reach X. If $\rho_C^{(k)} \geq r$ the sites are infeasible (Proposition 7), and C has to be enlarged, with depth leads or strips placed into the sulci, or a larger radius accepted. Otherwise add contacts one at a time, updating $\rho_1, \ldots, \rho_k$ exactly after each, until $\rho_k < r$: when sites may be anywhere on X, by farthest-point sampling; when they are

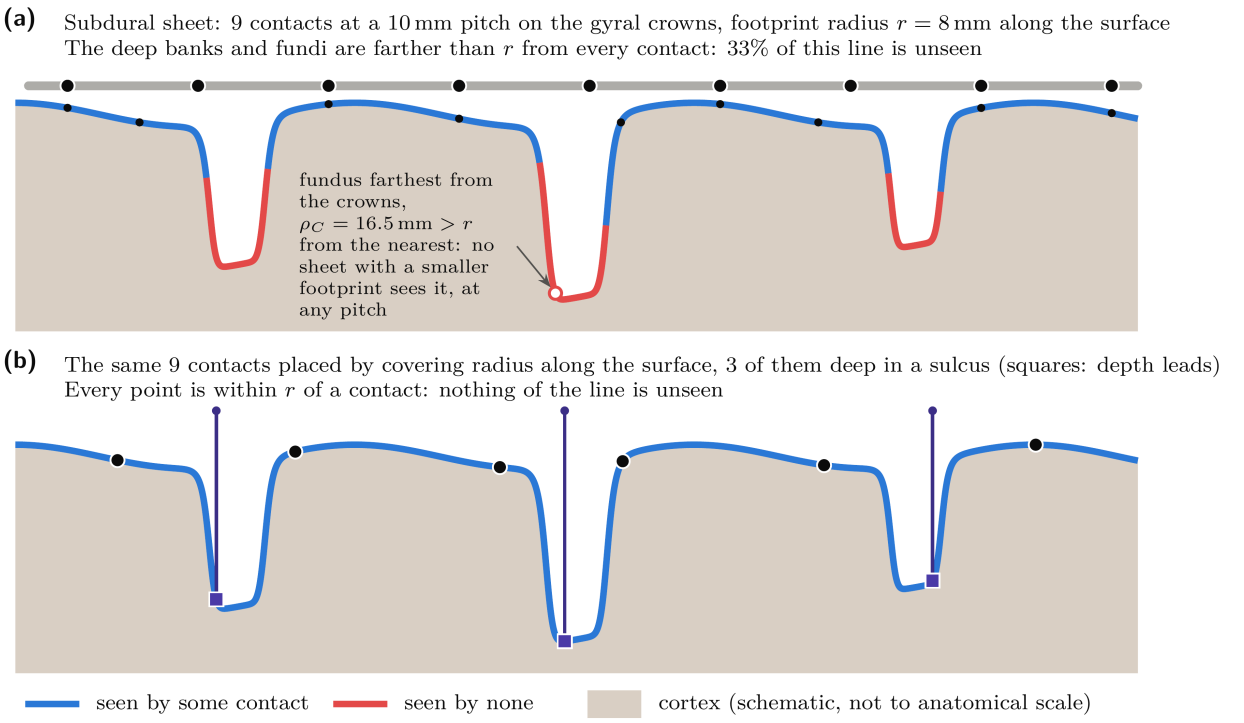


FIGURE 4. The site floor in one cross-section (schematic; distances are arc length along the pial line). (a) A subdural sheet with 9 contacts at a 10 mm pitch; large dots are contacts on the sheet and small dots the crown points they rest on. With footprint radius $r = 8$ mm the deep banks and fundi are seen by no contact, 33% of the line. The fundus marked is the point farthest from the crowns: its nearest crown point is $\rho_C = 16.5$ mm away, so no sheet with a footprint radius below that sees it, at any pitch (Proposition 7). (b) The same 9 contacts placed by covering radius along the line, 3 of them deep in a sulcus on depth leads: every point is within r of a contact. Geometry from `tikz/fold_geometry.py`.

restricted to C, by adding the site of C closest, in $e(c, T)$, to the worst covered triangle of X; by Proposition 2 the design then sees all of X k times, on the mesh and triangle by triangle. Then report the certificate of the covered region with its disk test, and the capture curves W_1, W_k and L. The procedure needs the patient's pial surface, which can be reconstructed from the presurgical MRI with standard tools [5, 12].

5. Results on a template cortex

Surface and territory. The surface is the left pial surface of the colin27 template [17], as distributed with FieldTrip [26]: 175 409 vertices and 350 814 triangles, every edge in exactly two triangles, Euler characteristic 2. The territory X is every triangle within 35 mm, along the surface, of a parietal anchor at MNI $(-58, -38, 38)$: 4977 vertices, 9659 triangles and 3777 mm^2, a topological disk. The reference array is the 8×8 parietal grid of the FieldTrip ECoG tutorial layout [33], 10 mm pitch, transferred to the template by laying the lattice on the envelope of the hemisphere and assigning each lattice point to the nearest gyral vertex; its median neighbour spacing is 9.75 mm in a straight line and 10.24 mm along the surface, and 49 of its 64 contacts lie outside X. This is template geometry, not a patient's implantation.

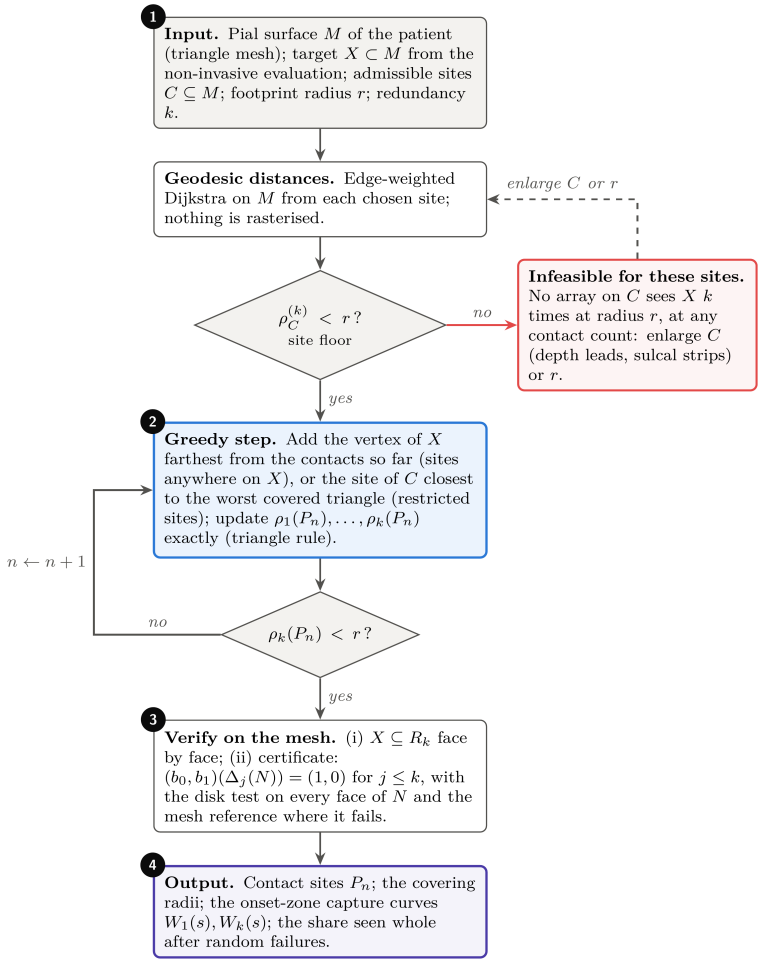


FIGURE 5. The design procedure. The site-floor test (Proposition 7, with the k-th nearest site) rules out, before any placement, admissible sites that cannot see X k times at radius r. Greedy steps add contacts until the k-th covering radius drops below r (Proposition 2); for $k = 1$ with sites anywhere on X they are within a factor two of optimal, up to one mesh edge (Proposition 9). The design is then checked on the mesh and its onset-zone capture is reported (Proposition 4).

Where the contacts sit decides the coverage. At $r = 8$ mm the 8×8 grid leaves 2082.0 mm^2 (55.1%) of X unseen, and its covered region, with Betti numbers $(b_0, b_1) = (9, 2)$, is in pieces and has holes. The same 64 contacts placed by farthest-point sampling over X leave 1.0 mm^2 unseen (Figs. 6 and 8b). At 10 and 12 mm the grid still misses 41.3% and 28.6%, and the placed array misses nothing. The covering radius says why (Fig. 7). The grid's is 21.96 mm. The placed array's is 8.12 mm, just above 8 mm, which is why 1.0 mm^2 stays unseen; 66 contacts bring it to 7.77 mm and leave nothing.

The site floor on real anatomy. Take C to be the gyral crowns, the vertices of negative sulcal depth, where a sheet on the envelope puts its contacts. A sheet is not confined to the crowns inside X, so we count every crown vertex within 25 mm of X, 7279 of them; none farther can see any triangle of X at the radii that matter here. The site floor is $\rho_C = 11.24$ mm, at a triangle whose centroid is at MNI $(-50.5, -44.4, 23.2)$, part way down a sulcal bank, and $\rho_C^{(2)} = 11.52$ mm for double coverage. (On the vertex rule the floor is 10.94 mm; counting only the 3016 crown vertices

(a) the published 8 × 8 grid, $r = 8$ mm (b) the same 64 contacts, placed by radius

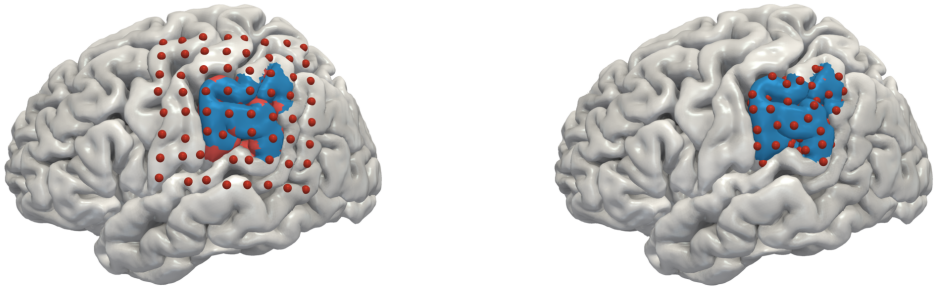


FIGURE 6. The target territory X on the colin27 left hemisphere at footprint radius $r = 8$ mm, blue where some contact sees the cortex and red where none does. (a) The documented 8×8 grid (red spheres), laid on the envelope of the hemisphere: most of the unseen cortex lies on sulcal banks, hidden from this view. (b) The same 64 contacts placed by covering radius. Rendered from the repository's result files with PyVista.

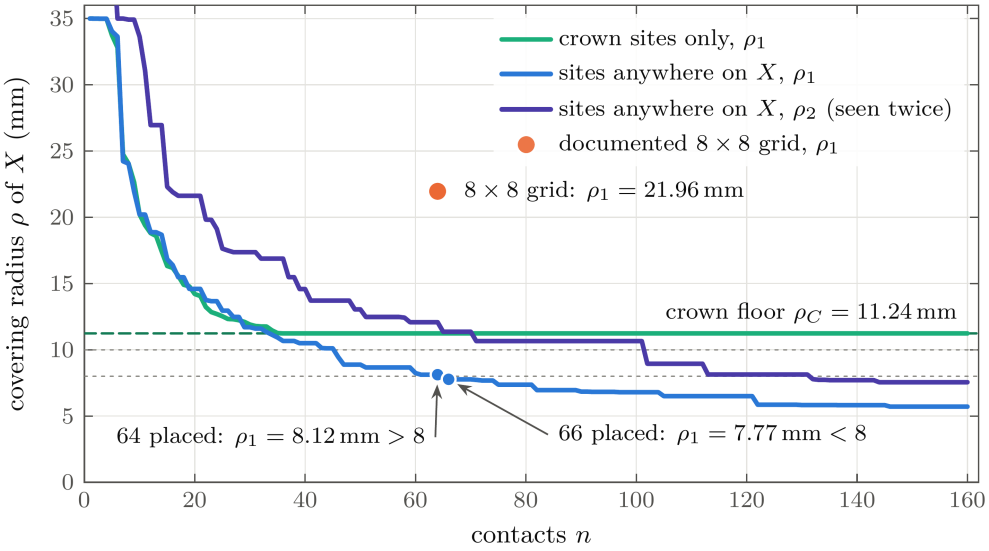


FIGURE 7. Covering radius of X against contact count, on the triangle rule. Blue: sites anywhere on X, placed by farthest-point sampling. Green: sites on gyral crowns within reach of X, which cannot fall below the crown floor $\rho_C = 11.24$ mm (dashed). Violet: the second covering radius ρ_2 of the blue sequence, which sets double coverage. Orange: the documented 8×8 grid. Dotted lines mark $r = 8$ and 10 mm; a design sees all of X at radius r exactly when its curve is below the line (Proposition 2).

inside X would give 12.22 mm.) A crown-only sequence, which at each step adds the crown site closest to the worst covered triangle, reaches the floor at $n = 36$ and stays there (Fig. 7). By Proposition 7, no array with its contacts on the crowns sees all of X at a footprint radius at or below this floor, whatever its pitch or contact count. With 64 crown contacts at $r = 8$ mm, 89.7 mm^2 of X stays unseen.

How few contacts. Table 2 gives, at each footprint radius, the fewest contacts of the farthest-point sequence that see all of X once and twice, found by scanning one contact at a time; at every radius,

TABLE 2. Fewest contacts of the farthest-point sequence that see all of X once ($k = 1$) and twice ($k = 2$), scanned one contact at a time, with their covering radii (triangle rule), the Betti numbers of R_k from the nerve and from the mesh, and the packing lower bound for single coverage (Proposition 8).

	$k = 1$				$k = 2$				
r (mm)	n	ρ_1 (mm)	nerve	mesh	n	ρ_2 (mm)	nerve	mesh	bound
14	22	13.75	$(1,0)$	$(1,0)$	41	13.71	$(1,0)$	$(1,1)$	8
12	29	11.70	$(1,0)$	$(1,0)$	65	11.36	$(1,0)$	$(1,1)$	8
10	46	9.45	$(1,0)$	$(1,0)$	102	8.95	$(1,0)$	$(1,0)$	12
9	47	8.88	$(1,0)$	$(1,0)$	102	8.95	$(1,0)$	$(1,1)$	13
8	66	7.77	$(1,0)$	$(1,0)$	132	7.78	$(1,0)$	$(1,0)$	16

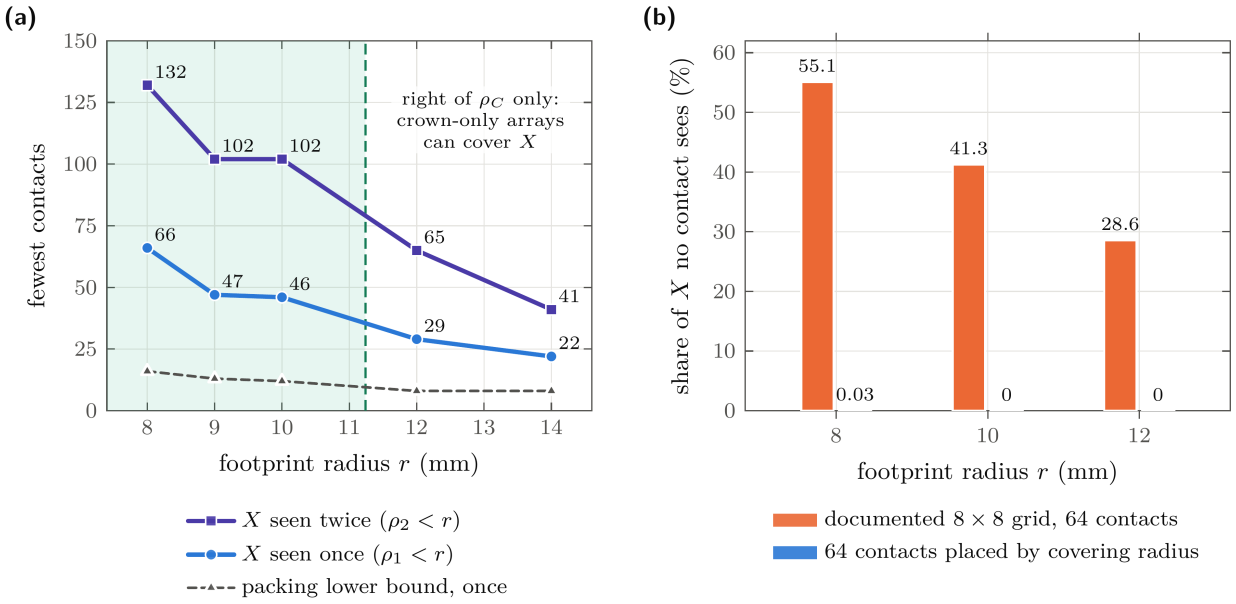


FIGURE 8. (a) Fewest contacts of the farthest-point sequence that see all of X once (blue) and twice (violet), against the packing lower bound for single coverage (grey). Left of the crown floor (shaded) no array with contacts on the crowns sees all of X at any count. (b) The share of X that no contact sees, for the documented 8×8 grid and for the same 64 contacts placed by covering radius.

one contact fewer leaves part of X unseen. Single coverage takes from 22 contacts at 14 mm to 66 at 8 mm, against packing lower bounds of 8 to 16, and double coverage from 41 to 132 (Fig. 8a). Every design below the crown floor needs sites off the crowns: between 27% and 33% of their sites lie in sulcal territory that a sheet cannot reach and a depth lead or a sulcal strip can. The certificate agrees with the mesh for every single-coverage design. For double coverage at 9, 12 and 14 mm the nerve gives $(1,0)$ for R_2 while the mesh finds a hole in it; since all of X is seen twice, the hole lies outside X, and it is missed by the nerve because geodesic footprints on a folded surface are not always convex. This is why containment is always decided by Proposition 2 on the mesh.

Seeing the onset zone whole. Table 3 and Fig. 9 give the capture of an onset zone of radius s placed uniformly in X. With the 8×8 grid at $r = 8$ mm, an onset zone of radius 5 mm is seen whole in 18.5% of its possible positions and missed entirely in 18.9%; one of radius 10 mm is seen whole in

TABLE 3. Covering radii and onset-zone capture of each design on X, on the triangle rule. W_1: positions in which an onset zone of radius s is seen whole; L: missed entirely; W_2: seen twice, hence whole after any one failure; h: radius of the largest onset zone that can be missed entirely. Percentages of admissible positions, area-weighted.

design	r	ρ_1	ρ_2	unseen	W_1 (%)		L (%)	h	W_2 (%)
	(mm)	(mm)	(mm)	(mm^2)	$s=5$	$s=10$	$s=5$	(mm)	$s=5$
8×8 grid, 64	8	21.96	25.02	2082.0	18.5	0.1	18.9	14.5	0.0
8×8 grid, 64	10	21.96	25.02	1559.4	33.2	4.5	10.1	12.6	2.8
crown sites, 64	8	11.24	12.67	89.7	82.6	55.1	0.0	2.9	6.4
placed, 64	8	8.12	12.08	1.0	98.1	95.6	0.0	1.3	3.5
placed, 66 ($k=1$)	8	7.77	11.36	0.0	100.0	100.0	0.0	0.0	3.6
placed, 132 ($k=2$)	8	5.83	7.78	0.0	100.0	100.0	0.0	0.0	100.0
placed, 46 ($k=1$)	10	9.45	13.71	0.0	100.0	100.0	0.0	0.0	4.2
placed, 102 ($k=2$)	10	6.79	8.95	0.0	100.0	100.0	0.0	0.0	100.0

0.1%, and one of radius up to 14.5 mm can lie in X and be missed entirely. The lower bound of Proposition 4(2), $\max_x \min\{\mathrm{d}_1(x,P)-r,\ \mathrm{d}(x,M\setminus X)\}$, evaluated at the vertices of X, is 13.57 mm. Sixty-four crown contacts never miss a 5 mm onset zone entirely but see it whole in only 82.6% of positions: the rest are seen in part, the case of truncated localisation. The same 64 contacts placed anywhere on X see it whole in 98.1% of positions, the remainder being the positions within 5 mm of the 1.0 mm^2 they leave unseen, where an onset zone smaller than 1.3 mm can even be missed. The 66 contacts that see all of X see every onset zone of every size whole, as Proposition 4(1) requires.

When contacts fail. Single coverage is fragile. The 66-contact design sees a 5 mm onset zone twice in only 3.6% of positions, so one well-placed failure can open a gap under most onset zones. Double coverage repairs this: 132 contacts at 8 mm and 102 at 10 mm see all of X twice, $\rho_2 = 7.78$ and 8.95 mm, and every onset zone survives any single failure.

Under random failure of q contacts (100 draws for each q), the share of 5 mm onset zones still seen whole falls, for the 66-contact single-coverage design, to 97.1% at $q=1$, 89.1% at $q=4$ and 78.6% at $q=8$ (5th percentile over draws 67.8%). For the double-coverage designs the lowest mean over all q is 99.8% at 8 mm and >99.9% at 10 mm (Fig. 10). The 8×8 grid starts at 18.5% and ends at 14.8%. The price of failure-proof sampling is contacts: 132 against 66 at 8 mm, and 102 against 46 at 10 mm, which is the trade the design procedure makes explicit.

6. Discussion

What optimal coverage buys. The results separate two things that "coverage" runs together: how many contacts an array has, and where they sit. On this territory the documented grid and a placed array with the same 64 contacts have covering radii of 21.96 and 8.12 mm, and differ by more than half the territory in what they see. The covering radius explains the difference in one number, predicts the largest onset zone an array can miss (Proposition 4), and is cheap enough to optimise directly. Pushing ρ_k below r gives a worst-case guarantee that no average-case measure gives: every onset zone of every size in X is seen whole, and stays so after any $k-1$ failures (Proposition 3).

The site floor explains why subdural sheets have limits that no pitch or contact count removes (Proposition 7). Between 27% and 33% of the contacts of the designs below the floor sit in sulcal territory, so an optimal design for a folded territory is in part a depth-lead design. Stereo-EEG and hybrid implantations reach sulcal and deep cortex that subdural grids do not [20, 27]. The procedure here says, for a given patient and territory, how much of the territory a sheet can see at all, before anything is implanted.

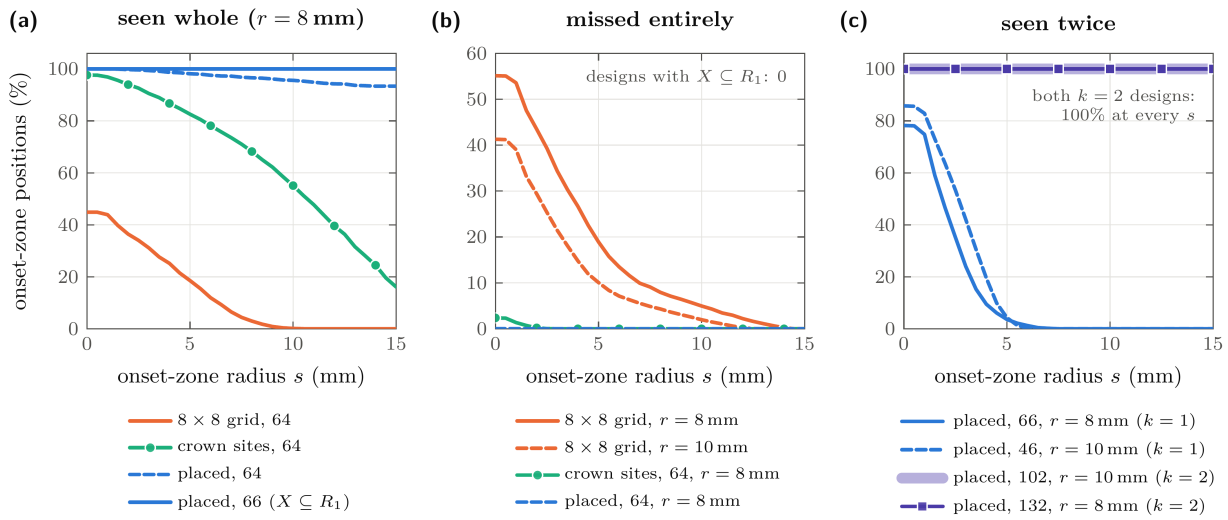


FIGURE 9. Onset-zone capture against onset-zone radius s, over all positions in which the zone fits in X (triangle rule; zones measured between triangle centroids). (a) Seen whole, $W_1(s)$, at $r = 8$ mm: the 8×8 grid, 64 contacts on the crowns, 64 and 66 contacts placed by covering radius. (b) Missed entirely, $L(s)$; designs that see all of X miss nothing, and 64 placed contacts can miss only onset zones smaller than 1.3 mm, too small to show. (c) Seen twice, $W_2(s)$, which by Proposition 3 is the share still seen whole after any one contact fails: single-coverage designs (blue) lose it quickly with s, and both double-coverage designs (violet) keep it at 100% for every s, so their lines coincide. Data from `run_soz_capture.py`.

What it does not buy. Seeing the onset zone whole is a condition for localising it, and localising it is a condition for resecting it. Neither is sufficient for seizure freedom. The recorded onset zone may not be the whole epileptogenic zone, the territory X may be wrong, and eloquent cortex may forbid the resection. Whether better coverage changes outcome is therefore an empirical question, and the published evidence does not settle it: the completeness of resection of the recorded onset zone was not associated with outcome in one cohort [13], while spatial perturbation of implantations has been proposed as a way to test whether the epileptogenic zone was sampled at all [19].

A test of clinical relevance. The test needs patients with four things together: the reconstructed pial surface, the implanted contact coordinates, the resection and the surgical outcome. Open intracranial datasets do not yet reliably provide all four. OpenNeuro ds003029 [24], for example, releases intracranial recordings with clinically annotated channels and surgical outcome, but no MRI or contact coordinates, which would have to come from the treating centres. For each patient, compute on the patient's own surface the covering radius of the resected territory by the implanted contacts, W_1 and L over it, and the same for the design this paper's procedure would have produced with the same number of contacts. Then test, under a plan fixed before any outcome label is read, whether coverage of the resected territory predicts seizure freedom. A null result would bound the clinical value of optimal coverage. A positive one would justify a prospective comparison of designed and conventional implantations.

Limitations. Everything here is computed on one template hemisphere and one territory, not on patients. The footprint radius r is a modelling parameter, the distance along the cortex over which a contact is credited with recording an onset; it is swept, not measured, and calibrating it from lead fields or from recorded spread is the most important extension. Geodesic distance is shortest-path

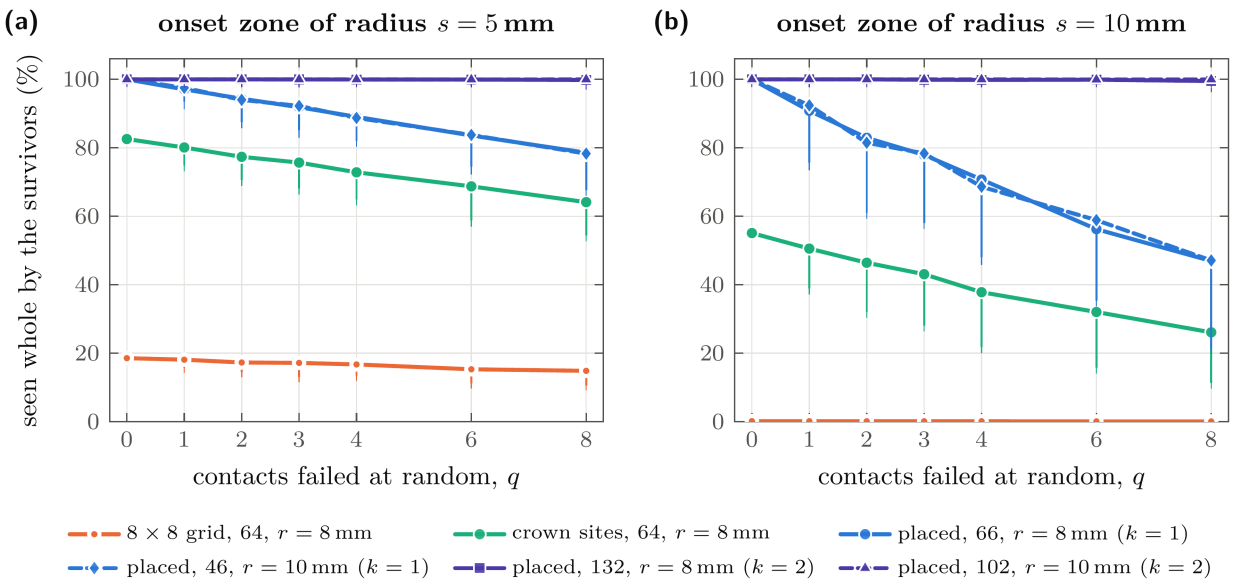


FIGURE 10. Onset zones still seen whole when q contacts fail at random, for onset zones of radius (a) 5 mm and (b) 10 mm. Lines are means over 100 random failure sets for each q; whiskers reach down to the 5th percentile. Single-coverage designs ($k = 1$, blue) degrade with every failure; double-coverage designs ($k = 2$, violet) stay near 100% through eight failures. The uniform failure model is an assumption; failures clustered in space would be harder on every design.

distance on the mesh edges, which slightly overestimates the true surface distance, and the triangle rule counts a triangle as seen only when one contact sees all of it; both make the designs conservative by at most about one mesh edge. Contact sites are mesh vertices, a site off the crowns stands for a depth or sulcal contact without modelling its trajectory, and surgical constraints (vessels, eloquent cortex, safe entry points) are not modelled; they enter the procedure as a smaller site set C, which Proposition 7 then prices. The uniform priors over onset position and over failures are assumptions. Failures clustered in space, as real contact failures on scalp arrays have been found to be [4], would be harder on every design, and would reward coverage deeper than $k = 2$.

7. METHODS

Surface and distances. The left pial surface of colin27 is taken from FieldTrip at commit `8e2307d`, in MNI millimetres; it is a de-identified reference anatomy, not patient data. Distances are Dijkstra distances [9] on the edge graph with Euclidean edge lengths. A triangle belongs to the footprint of a contact when all three of its vertices lie within distance r of it; the unseen area of X is the area of its triangles in no footprint. Gyral vertices are those of negative FreeSurfer sulcal depth.

Designs. Farthest-point sampling starts from the vertex of X nearest to the anchor and adds, at each step, a vertex of X farthest from the contacts so far. The triangle covering radii ρ_1 and ρ_2 of every prefix of this sequence are computed exactly from $e(p, T)$, and the fewest contacts of Table 2 are the first prefixes with $\rho_k < r$. The crown floor is computed from $e(c, T)$ for every crown vertex within 25 mm of X. The crown-only sequence adds, at each step, the crown site with the smallest $e(c, T^*)$ to the worst covered triangle T^* that is not yet at its own floor. Packing bounds come from a maximal $2r$-separated set of vertices of X. The certificate is computed with the nerve of the footprints, the Betti numbers of $\Delta_k(N)$ over GF(2), a disk test on every face of

the nerve, and a mesh reference for R_k computed without the nerve, all as in [4]. The designs are `run_ecog_designs.py`, about five minutes.

Capture and failure. Every triangle centroid is added to the edge graph as a node joined to the three vertices of its triangle. For each design, one multi-source Dijkstra run from the centroids of the triangles of X held by fewer than k footprints gives, at every centroid of X, the distance to $X \setminus R_k$; one from the centroids of the seen triangles gives the distance to R_1; and one from the centroids outside X decides admissibility. By Proposition 4, W_1, W_2 and L follow by thresholding at each s from 0 to 15 mm in steps of 0.5 mm, with triangle areas as the measure. For contact failure, $q \in \{1, 2, 3, 4, 6, 8\}$ contacts are removed uniformly at random, 100 times for each q with fixed seeds, the survivors' footprints are recomputed, and W_1 is taken again at $s = 5$ and 10 mm. The analysis is `run_soz_capture.py`, about a minute.

Lean formalization. The combinatorial content of the propositions on the mesh is proved in the Lean 4 proof assistant [7], in the directory `lean/` of the repository (core Lean 4.22.0 without libraries, built by `lake build`). Contacts and triangles are elements of arbitrary types, and the distances $e(p, T)$ and d are functions to the natural numbers, for instance in micrometres. Only their order is used, together with the symmetry and triangle inequality of d for Propositions 8 and 9. The theorems are:

- Proposition 2 (`cover_iff`): every triangle of X lies in at least $k \geq 1$ footprints if and only if, for every triangle, the k-th smallest of its distances $e(p, T)$ is below r;
- Proposition 3 for distinct contacts (`failure_proof`, `zone_failure_proof`);
- the equivalence behind Proposition 4 on the mesh: an onset zone about a centroid consists of triangles of a given kind exactly when every other triangle of X is farther than s from that centroid (`capture_by_erosion`);
- Proposition 7, in its k-fold form (`site_floor`, `site_floor_k`);
- Proposition 8 (`packing`);
- the pigeonhole step of Proposition 9 (`farthest_point_core`);
- the per-contact bounds $\mathrm{d}(v, p) \leq e(p, T) \leq \mathrm{d}(v, p) + \ell$ for a vertex v of T, behind $\rho_k^{\mathrm{v}} \leq \rho_k \leq \rho_k^{\mathrm{v}} + \ell$ (`le_triDist`, `triDist_le`).

Every theorem depends only on the standard axioms of Lean (propositional extensionality, choice and the soundness of quotients), and no proof is left open. Not formalized are the statements on the continuous surface, among them the measure argument of Proposition 4(2) and its lower bound on $h(P)$, and the separation property of the farthest-point sequence to which the pigeonhole step is applied.

Figures. Figures 1–5 and 7–10 are TikZ and pgfplots sources, drawn from the result files by `make_ecog_paper_data.py` and rendered at 600 dpi; the colours are a single palette checked for colour-blind separation, with markers and line styles as a second channel. Figure 6 is the PyVista render of the repository.

Data availability

No patient data are used. The template surface, the designs (every contact by mesh vertex and MNI coordinate) and all result files are in the repository below.

Code availability

All code, result files and figure sources are in the GitHub repository `https://github.com/creelie/kshadow-coverage` [4]. The Python code of this paper (`run_ecog_designs.py`, `run_soz_capture.py` and the designs and certificates they build on), its result files and its figure sources were introduced in release v1.5.0, archived on Zenodo at doi:10.5281/zenodo.22953062. The Lean formalization in `lean/` was added to the repository after that release. This revision additionally

includes exact disk predicates, whole-face proofs and rigorous path bounds in release v1.6.0 at `https://github.com/MeBose007/kshadow-coverage/releases/tag/v1.6.0`. The manuscript is `paper/ecog/ecog_coverage.tex`, built by `paper/ecog/build.sh`. Every number in it is a macro generated from the result files.

Acknowledgements

The Python computations and the Lean 4 formalization of this paper were carried out with the assistance of Claude (Anthropic). They use the code in the GitHub repository `https://github.com/creelie/kshadow-coverage`, release v1.5.0, archived on Zenodo at doi:10.5281/zenodo.22953062.

The v1.6.0 exact-geometry extension, continuous-face proof, integer-path audit and manuscript revision were prepared with Codex assistance under the direction of Deep Bhattacharjee. The new analytic whole-face theorem and path-arithmetic implementation have written proofs and Python checks; they are not included in the earlier Lean formalization.

Author contributions

The idea of certifying k-fold sensor coverage through iterated nerve complexes was first developed by D.B. with U.B. [2]. A.B. extended it to electrode arrays on the cortex. D.B., U.B. and A.B. wrote the code and carried out the analyses together, and all three authors wrote the paper.

Funding

None.

Competing interests

None.

References

1. Fabrice Bartolomei, Stanislas Lagarde, Fabrice Wendling, et al., *Defining epileptogenic networks: contribution of SEEG and signal analysis*, Epilepsia **58** (2017), no. 7, 1131–1147.
2. Deep Bhattacharjee and Ushashi Bhattacharya, *Iterated nerve complexes for k-fold sensor coverage*, Preprints 2026, 2026061351 (version 2), 2026.
3. Karol Borsuk, *On the imbedding of systems of compacta in simplicial complexes*, Fundamenta Mathematicae **35** (1948), 217–234.
4. Aditi Bose, Deep Bhattacharjee, and Ushashi Bhattacharya, *Supplementary code for: Topological Certificates of Redundant Coverage for Cortical Electrode Arrays, version 1.5.0*, `https://github.com/creelie/kshadow-coverage`, 2026, Release v1.5.0 archived on Zenodo, doi:10.5281/zenodo.22953062.
5. Anders M. Dale, Bruce Fischl, and Martin I. Sereno, *Cortical surface-based analysis I: segmentation and surface reconstruction*, NeuroImage **9** (1999), no. 2, 179–194.
6. Elena De Momi, Chiara Caborni, Francesco Cardinale, et al., *Multi-trajectories automatic planner for StereoElectroEncephaloGraphy (SEEG)*, International Journal of Computer Assisted Radiology and Surgery **9** (2014), no. 6, 1087–1097.
7. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
8. Vin de Silva and Robert Ghrist, *Coverage in sensor networks via persistent homology*, Algebraic & Geometric Topology **7** (2007), 339–358.
9. Edsger W. Dijkstra, *A note on two problems in connexion with graphs*, Numerische Mathematik **1** (1959), 269–271.
10. Jerome Engel, Jr., Michael P. McDermott, Samuel Wiebe, et al., *Early surgical therapy for drug-resistant temporal lobe epilepsy: a randomized trial*, JAMA **307** (2012), no. 9, 922–930.
11. Jerome Engel, Jr., Paul C. Van Ness, Theodore B. Rasmussen, and Lorna M. Ojemann, *Outcome with respect to epileptic seizures*, Surgical Treatment of the Epilepsies (Jerome Engel, Jr., ed.), Raven Press, New York, 2nd ed., 1993, pp. 609–621.
12. Bruce Fischl, *FreeSurfer*, NeuroImage **62** (2012), no. 2, 774–781.
13. Sarah J. Gascoigne, Nathan Evans, Gerard Hall, et al., *Incomplete resection of the intracranial electroencephalographic seizure onset zone is not associated with postsurgical outcomes*, Epilepsia **65** (2024), no. 9.

14. Teofilo F. Gonzalez, *Clustering to minimize the maximum intercluster distance*, Theoretical Computer Science **38** (1985), 293–306.
15. Hajo M. Hamer, Harold H. Morris, Edward J. Mascha, et al., *Complications of invasive video-EEG monitoring with subdural grid electrodes*, Neurology **58** (2002), no. 1, 97–103.
16. Dorit S. Hochbaum and David B. Shmoys, *A best possible heuristic for the k-center problem*, Mathematics of Operations Research **10** (1985), no. 2, 180–184.
17. Colin J. Holmes, Rick Hoge, Louis Collins, Roger Woods, Arthur W. Toga, and Alan C. Evans, *Enhancement of MR images using registration for signal averaging*, Journal of Computer Assisted Tomography **22** (1998), no. 2, 324–333.
18. Chi-Fu Huang and Yu-Chee Tseng, *The coverage problem in a wireless sensor network*, Mobile Networks and Applications **10** (2005), no. 4, 519–528.
19. Kassem Jaber, Tamir Avigdor, Daniel Mansilla, et al., *A spatial perturbation framework to validate implantation of the epileptogenic zone*, Nature Communications **15** (2024), 5253.
20. Prasanna Jayakar, Jean Gotman, A. Simon Harvey, et al., *Diagnostic utility of invasive EEG for epilepsy surgery: indications, modalities, and techniques*, Epilepsia **57** (2016), no. 11, 1735–1747.
21. Barbara C. Jobst and Gregory D. Cascino, *Resective epilepsy surgery for drug-resistant focal epilepsy: a review*, JAMA **313** (2015), no. 3, 285–293.
22. Patrick Kwan, Alexis Arzimanoglou, Anne T. Berg, Martin J. Brodie, W. Allen Hauser, Gary Mathern, Solomon L. Moshé, Emilio Perucca, Samuel Wiebe, and Jacqueline French, *Definition of drug resistant epilepsy: consensus proposal by the ad hoc Task Force of the ILAE Commission on Therapeutic Strategies*, Epilepsia **51** (2010), no. 6, 1069–1077.
23. Patrick Kwan and Martin J. Brodie, *Early identification of refractory epilepsy*, New England Journal of Medicine **342** (2000), no. 5, 314–319.
24. Adam Li, Chester Huynh, Zachary Fitzgerald, et al., *Neural fragility as an EEG marker of the seizure onset zone*, Nature Neuroscience **24** (2021), no. 10, 1465–1474.
25. Jeffrey P. Mullin, Michael Shriver, Soha Alomar, et al., *Is SEEG safe? A systematic review and meta-analysis of stereo-electroencephalography-related complications*, Epilepsia **57** (2016), no. 3, 386–401.
26. Robert Oostenveld, Pascal Fries, Eric Maris, and Jan-Mathijs Schoffelen, *FieldTrip: open source software for advanced analysis of MEG, EEG, and invasive electrophysiological data*, Computational Intelligence and Neuroscience **2011** (2011), 156869.
27. Josef Parvizi and Sabine Kastner, *Promises and limitations of human intracranial electroencephalography*, Nature Neuroscience **21** (2018), no. 4, 474–483.
28. Felix Rosenow and Hans Lüders, *Presurgical evaluation of epilepsy*, Brain **124** (2001), no. 9, 1683–1700.
29. Donald R. Sheehy, *A multicover nerve for geometric inference*, Proceedings of the 24th Canadian Conference on Computational Geometry (CCCG), 2012.
30. Marc W. Slutzky, Luke R. Jordan, Todd Krieg, Ming Chen, David J. Mogul, and Lee E. Miller, *Optimal spacing of surface electrode arrays for brain–machine interface applications*, Journal of Neural Engineering **7** (2010), no. 2, 026004.
31. Rachel Sparks, Gergely Zombori, Roman Rodionov, et al., *Automated multiple trajectory planning algorithm for the placement of stereo-electroencephalography (SEEG) electrodes in epilepsy treatment*, International Journal of Computer Assisted Radiology and Surgery **12** (2017), no. 1, 123–136.
32. Susan S. Spencer, *Neural networks in human epilepsy: evidence of and implications for treatment*, Epilepsia **43** (2002), no. 3, 219–227.
33. Arjen Stolk, Sandon Griffin, Roemer van der Meij, Callum Dewar, Ignacio Saez, Jack J. Lin, Giovanni Piantoni, Jan-Mathijs Schoffelen, Robert T. Knight, and Robert Oostenveld, *Integrated analysis of anatomical and electrophysiological human intracranial data*, Nature Protocols **13** (2018), 1699–1723.
34. Samuel Wiebe, Warren T. Blume, John P. Girvin, and Michael Eliasziw, *A randomized, controlled trial of surgery for temporal-lobe epilepsy*, New England Journal of Medicine **345** (2001), no. 5, 311–318.
35. Heinz G. Wieser, Warren T. Blume, David Fish, et al., *ILAE Commission Report. Proposal for a new classification of outcome with respect to epileptic seizures following epilepsy surgery*, Epilepsia **42** (2001), no. 2, 282–286.

Centre for Computational Natural Sciences and Bioinformatics, International Institute of Information Technology, Hyderabad, India
URL: https://orcid.org/0000-0002-3036-8223

Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India
Email address: itsdeep@live.com
URL: https://orcid.org/0000-0003-0466-750X

Formerly, National Taiwan University, Taipei 106319, Taiwan
Email address: rai.bhattacharya.in@gmail.com
URL: https://orcid.org/0009-0002-2254-3914